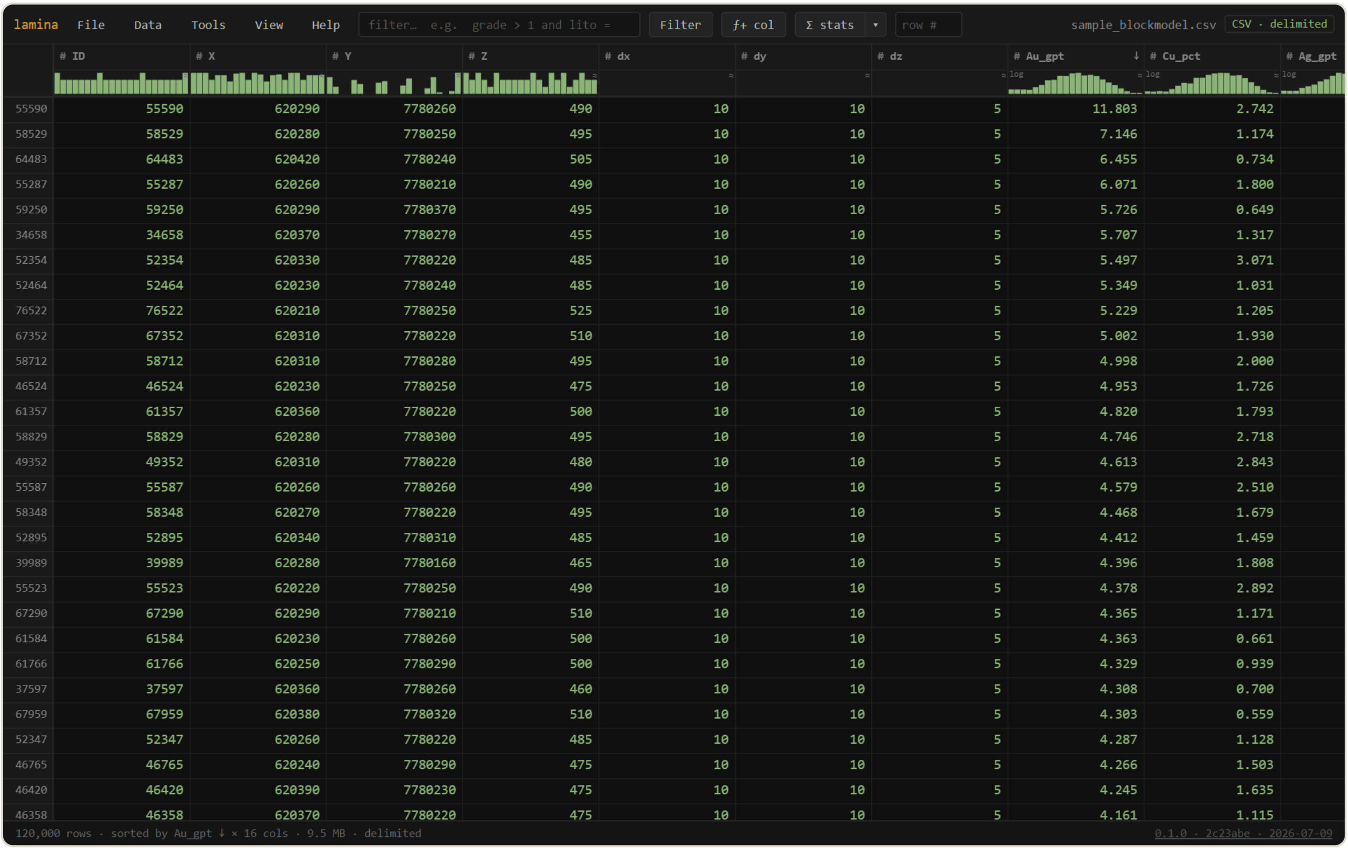
lamina

Open any file, however large — scroll, filter, sort, derive, summarize, and quality-check it. Windowed, read-only, offline.

🔒 **Sealed** — no network, nothing leaves your machine · single self-contained HTML file · download lamina.html · open the app ↗

User documentation · Auditable / Geoscientific Chaos Union · gentropic.org/lamina

lamina never loads the whole file, so size isn't the problem: a multi-gigabyte CSV or Datamine `.dm` scrolls in about a second, filters and sorts in place, and reports grade–tonnage — all without an import step, a server, or a byte leaving your machine.



	# ID	# X	# Y	# Z	# dx	# dy	# dz	# Au_gpt ↓	# Cu_pct	# Ag_gpt
55590	55590	620290	7780260	490	10	10	5	11.803	2.742	
58529	58529	620280	7780250	495	10	10	5	7.146	1.174	
64483	64483	620420	7780240	505	10	10	5	6.455	0.734	
55287	55287	620260	7780210	490	10	10	5	6.071	1.800	
59250	59250	620290	7780370	495	10	10	5	5.726	0.649	
34658	34658	620370	7780270	455	10	10	5	5.707	1.317	
52354	52354	620330	7780220	485	10	10	5	5.497	3.071	
52464	52464	620230	7780240	485	10	10	5	5.349	1.031	
76522	76522	620210	7780250	525	10	10	5	5.229	1.205	
67352	67352	620310	7780220	510	10	10	5	5.002	1.930	
58712	58712	620310	7780280	495	10	10	5	4.998	2.000	
46524	46524	620230	7780250	475	10	10	5	4.953	1.726	
61357	61357	620360	7780220	500	10	10	5	4.820	1.793	
58829	58829	620280	7780300	495	10	10	5	4.746	2.718	
49352	49352	620310	7780220	480	10	10	5	4.613	2.843	
55587	55587	620260	7780260	490	10	10	5	4.579	2.510	
58348	58348	620270	7780220	495	10	10	5	4.468	1.679	
52895	52895	620340	7780310	485	10	10	5	4.412	1.459	
39989	39989	620280	7780160	465	10	10	5	4.396	1.808	
55523	55523	620220	7780250	490	10	10	5	4.378	2.892	
67290	67290	620290	7780210	510	10	10	5	4.365	1.171	
61584	61584	620230	7780260	500	10	10	5	4.363	0.661	
61766	61766	620250	7780290	500	10	10	5	4.329	0.939	
37597	37597	620360	7780260	460	10	10	5	4.308	0.700	
67959	67959	620380	7780320	510	10	10	5	4.303	0.559	
52347	52347	620260	7780220	485	10	10	5	4.287	1.128	
46765	46765	620240	7780290	475	10	10	5	4.266	1.503	
46420	46420	620390	7780230	475	10	10	5	4.245	1.635	
46358	46358	620370	7780220	475	10	10	5	4.161	1.115	

A block model open in lamina. Every column header carries a live distribution glyph — histogram for numbers, top-values bar for categories — and the grid scrolls the file windowed, never resident.

Contents

1 What lamina is

lamina is a single self-contained HTML file — a **windowed data viewer** built for the files resource geology actually produces. It reads over a coarse block index, decoding only what's on screen (plus what a scan is currently sweeping), so the first look is near-instant at any size and memory stays flat.

It reads:

- **Delimited tables** — CSV / TSV / TXT / DAT, any size; GSLIB / Geo-EAS and whitespace dumps detected, `#` comment preambles skipped
- **Datamine `.dm`** — opened directly at any size, decoded on the fly (no conversion), with the same filter / sort / stats as CSV; filters read only the columns they need (strided), which is how a 17-column file filters 30× faster than a naïve decode
- **Archives** — zip / tar / gz / zst / bz2: peek inside and window huge compressed entries *without unpacking*
- **Text** → lines · **binary** → hex — nothing refuses to open

Everything is **read-only**: lamina never modifies your file. Derived things — calculated columns, filters, formats — live in the view (and can be saved as a lens), while Export writes a new file.

> **Tip —** No file handy? The empty screen offers ▸ *open a sample block model* — 120,000 rows generated locally, nothing downloaded.

2 Getting started

File → Open (`Ctrl+O`) or drag a file anywhere onto the window. Recent files are remembered (toggleable) and reopen with one click.

If a file reads wrong

Click the **kind badge** (top right) or **Data → Interpretation…** to force the delimiter, header on/off, rows to skip, comment prefix, decimal point vs comma, or the **character encoding** (UTF-8 · Windows-1252 · Latin-1 · UTF-16). A `⚠ encoding?` hint appears in the footer when bytes don't decode cleanly as UTF-8 — mojibake never passes silently.

3 The grid

The grid is windowed: scrolling a 1.8-million-row file reads just the rows in view. **Click a header** to sort (ascending → descending → off; sorts can stack across columns), the **row #** box jumps, and `Ctrl+F` opens **find** (case / whole-cell / regex / per-column scope / match count). Selected cells copy as TSV with `Ctrl+C`.

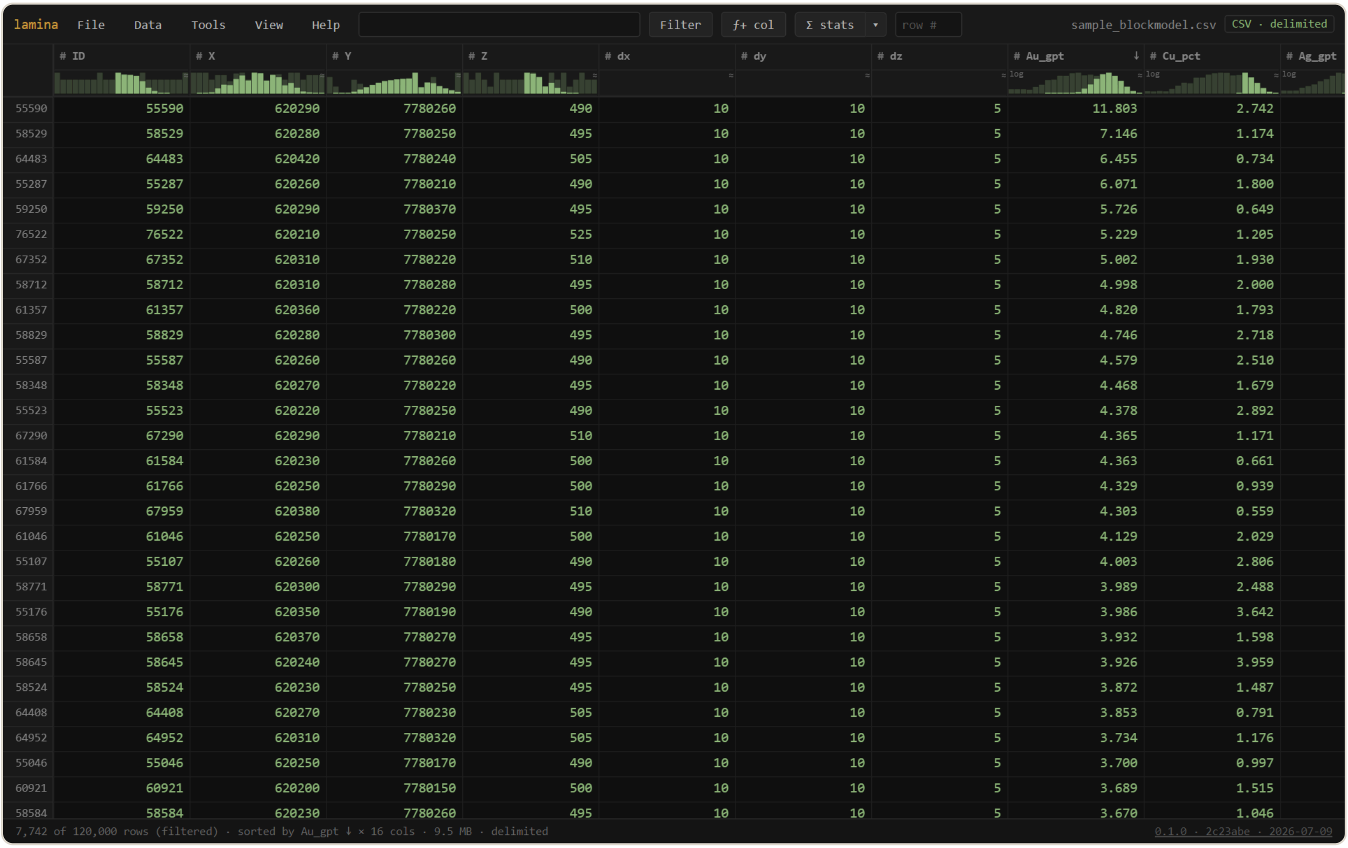
Two dockable panels (View menu): the **Columns** panel — search, show/hide, drag-reorder, pin-freeze, and per-column settings — and the **Record inspector** — one row's fields as a list that follows the selection; `↑`/`↓` step rows and ⊙ **pin** holds one for a side-by-side compare.

Right-click a numeric header → **Color scale** heatmaps the cells (8 palettes, linear/log, clip outliers, reverse) — patterns in a million rows read at a glance. **Number format** and **treat as text / number** live in the same menu.

4 Column distributions

Each column header carries a live **distribution glyph**: a histogram for numbers (with a null-rate bar; `≈` means sampled, `log` means auto log-scaled for skewed grades), a coloured top-values bar for categories. They're instruments, not decoration:

- **hover** — read the value / category under the cursor
- **click** — filter to that bin or category
- **drag** — brush a range → a filter (auto-applies when cheap, stages for `Enter` on huge files — Data → Gutter brush)
- **double-click** — the column's full Statistics



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55587	55587	620260	7780260	490	10	10	5	4.579	2.510	
58348	58348	620270	7780220	495	10	10	5	4.468	1.679	
55523	55523	620220	7780250	490	10	10	5	4.378	2.892	
67290	67290	620290	7780210	510	10	10	5	4.365	1.171	
61584	61584	620230	7780260	500	10	10	5	4.363	0.661	
61766	61766	620250	7780290	500	10	10	5	4.329	0.939	
67959	67959	620380	7780320	510	10	10	5	4.303	0.559	
61046	61046	620250	7780170	500	10	10	5	4.129	2.029	
55107	55107	620260	7780180	490	10	10	5	4.003	2.806	
58771	58771	620300	7780290	495	10	10	5	3.989	2.488	
55176	55176	620350	7780190	490	10	10	5	3.986	3.642	
58658	58658	620370	7780270	495	10	10	5	3.932	1.598	
58645	58645	620240	7780270	495	10	10	5	3.926	3.959	
58524	58524	620230	7780250	495	10	10	5	3.872	1.487	
64408	64408	620270	7780230	505	10	10	5	3.853	0.791	
64952	64952	620310	7780320	505	10	10	5	3.734	1.176	
55046	55046	620250	7780170	490	10	10	5	3.700	0.997	
60921	60921	620200	7780150	500	10	10	5	3.689	1.515	
58584	58584	620230	7780260	495	10	10	5	3.670	1.046	

Filtering: an expression in the box — or a drag across a header glyph — and the grid shows only the matches, windowed as ever. The footer counts them.

5 Filter

The filter box speaks a **SQL-WHERE**-style language: `grade > 1 and lito = "OXIDE"`. `Enter` applies, `Esc` clears. Highlights: `between … and …`, `in (…)`, `contains` / `like` / `matches` for text, `is blank` / `is filled`, functions (`round sqrt log if(c,a,b) bin coalesce isnum`…), and parentheses. Bare words are *columns* (so `fe > cu` compares two columns); quoted words are text. Blanks behave sanely — no SQL `NULL` traps. Autocomplete offers columns, functions, and data values.

On a `.dm` file the filter reads *only the columns the expression uses* — the strided fast path — so a selective filter over a huge model finishes in seconds.

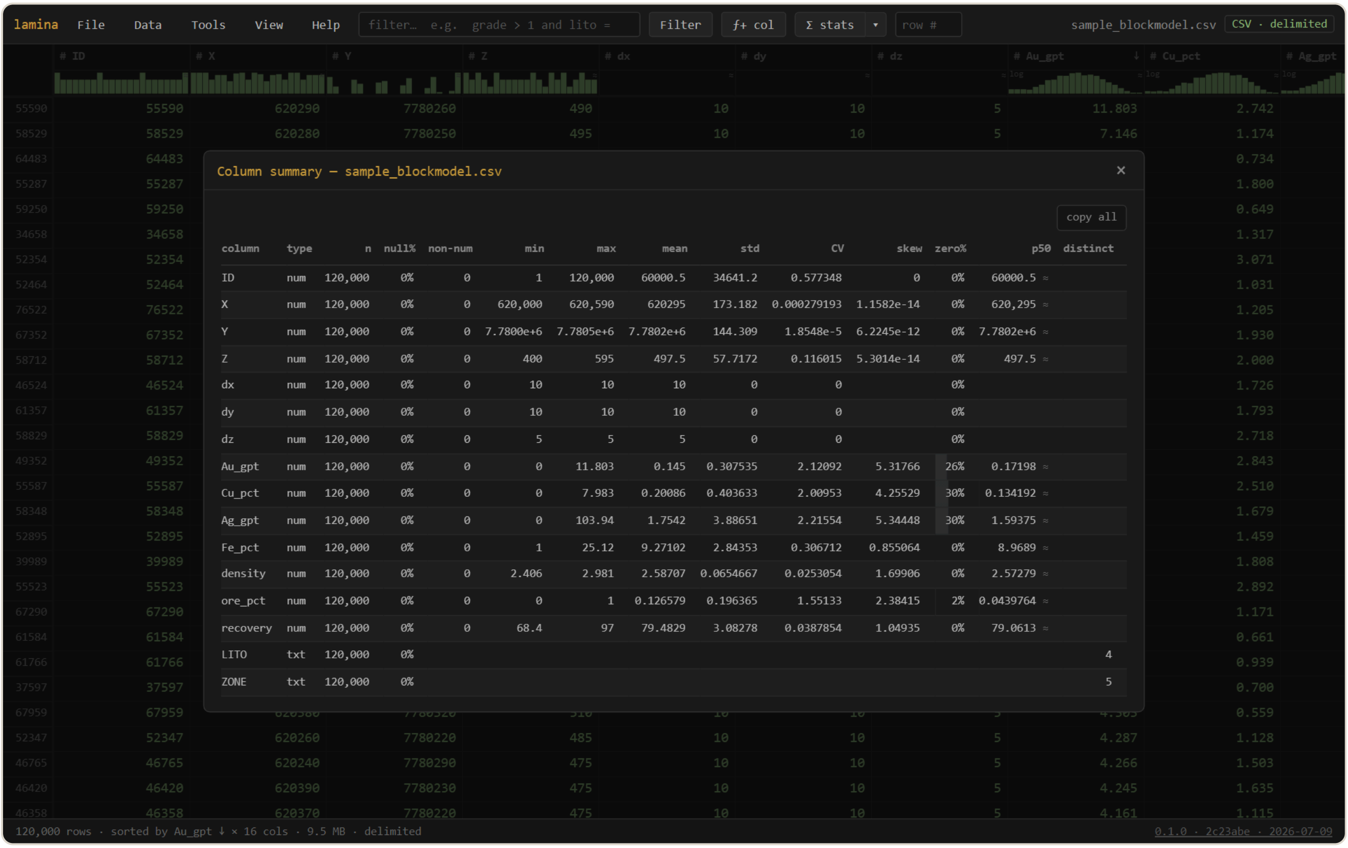
6 Calculated columns

The **ƒ+ col** button (or a header's right-click, or **Data** → **Calculated columns…**) adds a derived column from a formula in the same language as the filter: `grade * density`, `if(au > 1, "ore", "waste")`, `bin(rl, 10)` for elevation bands. Computed on the fly, never written; filter, sort, stat, group, and export it like any column.

7 Statistics

One column: double-click its glyph (or right-click → Statistics) — histogram with a log-x toggle, quantiles, count / nulls / non-numeric (with examples), min / max / mean / std / **CV** / **skew** / zeros, and **exclude zeros / negatives** re-stats for grade-only views. Copies as TSV.

Every column: Σ stats on the toolbar sweeps the whole table once and opens the **Column summary** — a sortable row per column (n · null% · non-num · min · max · mean · std · CV · skew · zero% · p50). Sort by **null%** or **non-num** to triage a new file in seconds; click a row for detail, click a name to jump to the column in the grid.



Σ stats — the whole file, one pass, one sortable table. The fastest way to meet a file you've never seen.

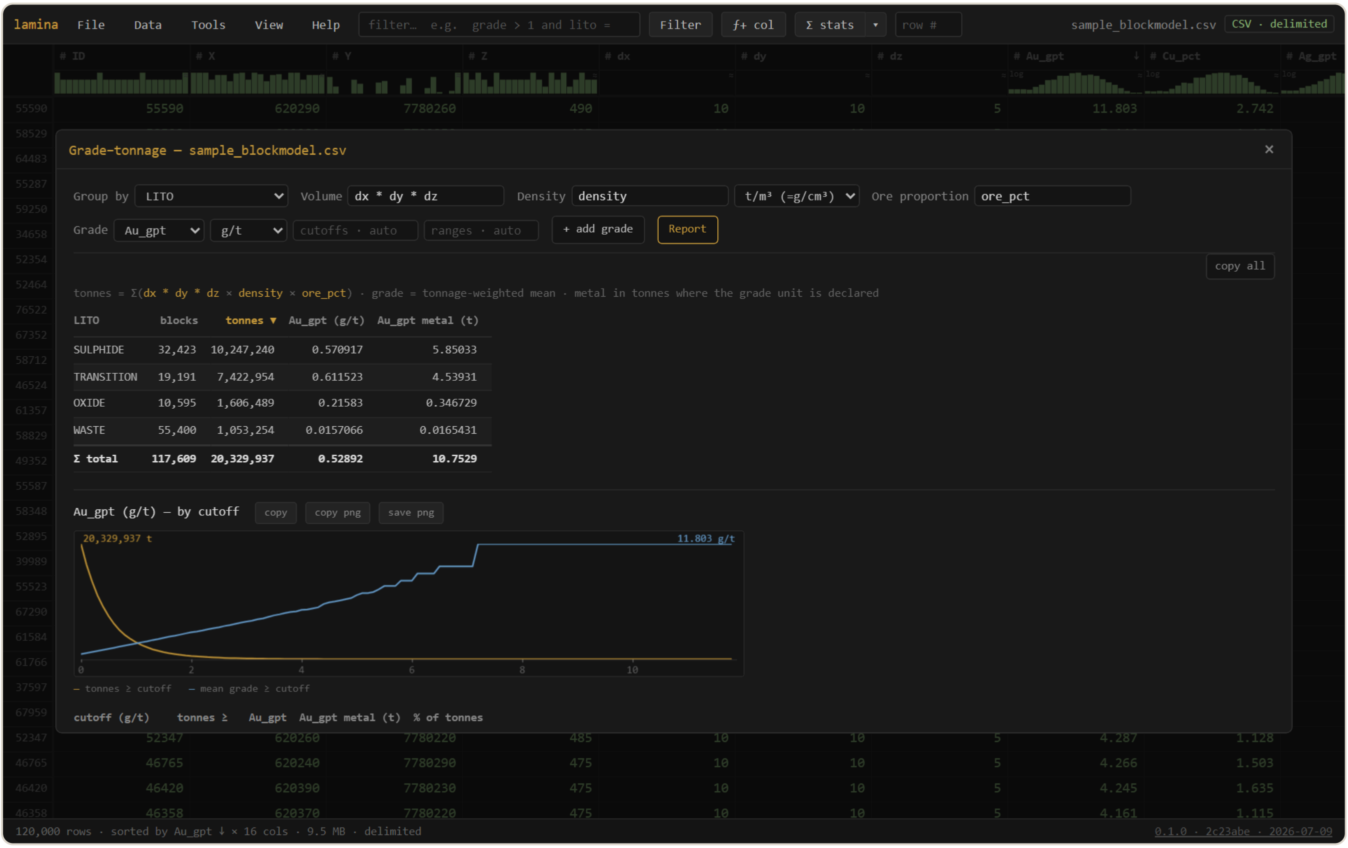
8 Group by

Tools → Group by… aggregates value columns per group, with an optional **weight** column — the weighted mean shows next to the plain one, so an unweighted-vs-length-weighted grade difference is visible instead of hidden. Click a group to filter the grid to it; copy as TSV. Numeric bands: add `bin(rl, 10)` as a calculated column and group by it.

9 Grade–tonnage

Tools → Grade–tonnage… is the resource view. **Volume · density · ore proportion** are expressions (a column, a constant, `dx * dy * dz` — auto-detected where obvious; blank = 1), with a **declared density unit** so a kg/m^3 column doesn't inflate tonnage a thousandfold. The report gives **tonnes · tonnage-weighted grade · contained metal** per domain (+ Σ total), and every grade field gets its **cutoff curve** — tonnes ≥ cutoff falling, mean grade rising — with a table at automatic round cutoffs *or your own* (`0.5, 1, 2` — computed exactly, not snapped).

- **Declared grade units** (%, g/t, ppm, oz/t) label the report and make *metal come out in tonnes*.
- **Manual plot ranges** per grade — `x:0..2 t:5e6 g:3` pins the cutoff window and the y-scales, so this month's curve overlays last month's at the same scale.
- Each curve offers **copy** (TSV) · **copy png** · **save png**.
- Respects the current filter, and computes windowed — a multi-GB model reports without loading.



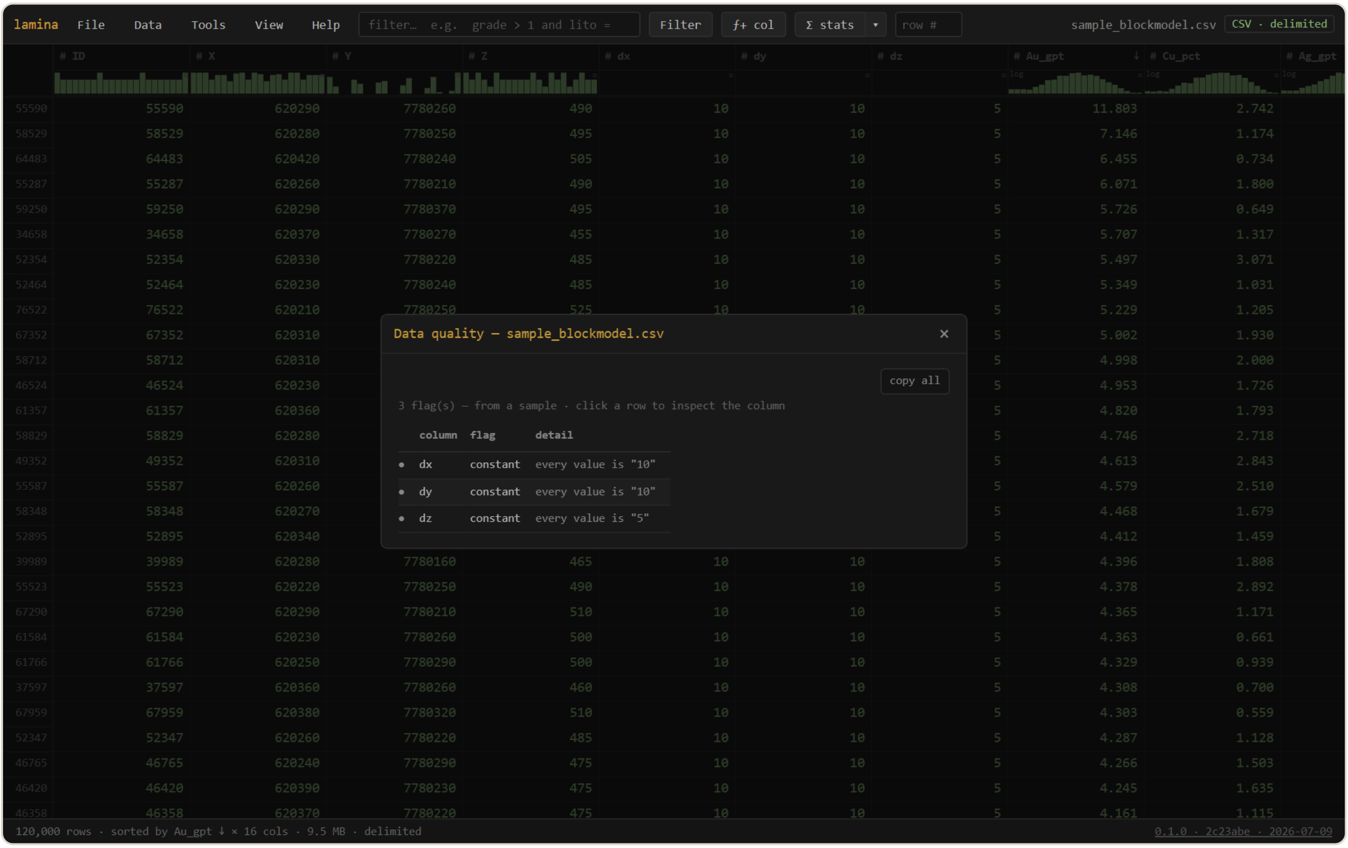
Grade–tonnage by domain with cutoff curves — tonnes falling, mean grade rising. Straight off the raw file, respecting the active filter.

10 Grid summary

Tools → Grid summary... infers a block model's **grid** from its centroid columns: origin · block size · count per axis, coordinate **ordering** (fastest/slowest), **sub-block detection**, parent-cell count and **fill %**. "No regular grid" is itself an answer — scattered data, or a broken export. Copy as TSV.

11 Data quality

Tools → Data quality… scans for the quiet bugs that bite an estimate: **leading zeros lost** (a code like `007` read as a number — with a one-click *fix: treat as text*), **non-numeric values** hiding in numeric columns, **missing-value sentinels** (`-999` …), thousands separators, whitespace padding, all-blank or constant columns, dates-as-text. Click a flag for that column's Statistics; `✓` when clean.



Data quality — the pre-flight check. Every flag is a thing that has, somewhere, quietly wrecked an estimate.

12 Lenses

Data → Save lens... saves your *view* — filter · sort · calculated columns · number formats · color scales · hidden columns — as a small `.lamina` file. Not the data: the recipe for looking at it. **Apply lens...** (or just open a `.lamina`) re-applies it to the current file, matching columns **by name**; anything that doesn't match is skipped and said. One setup, reused across every export with the same schema — and shareable with a colleague.

13 Export

Data → Export… writes the current view — filtered, sorted, calculated columns included, chosen columns only — to CSV/TSV, **streamed** straight to the file you pick: never uploaded, never fully held in memory. Opening a `.dm` and exporting *is* the `.dm`→CSV converter, at any size.

14 Security: Sealed

lamina is **Sealed**: it makes **no network requests** — a browser security policy (`connect-src 'none'`) blocks every connection, so there is no telemetry, no cloud, no auto-update. It is also **WASM-free** with the runtime in the clear: the code is readable and auditable rather than opaque binary. Your data — models, assays, exports — stays on your device, and its security artifacts (capability declaration, SHA-256, SBOM) are build-enforced, not asserted.

Verify the build and read how to bring it through approval at gentropic.org/security.

15 **Install & offline**

lamina runs at gentropic.org/lamina — install it as an app from the browser (it's also on the **Microsoft Store** as *GCU Lamina*), or download `lamina.html`: one file, no installer, that you can keep, e-mail, drop on a shared drive, or run fully offline by double-clicking. Nothing to set up, nothing that phones home. Works in current Chrome / Edge / Firefox.

A Reference

Keyboard & mouse

Input	Action
`Ctrl+O`	open a file
`Ctrl+F`	find in the table — `Enter` / `Shift+Enter` next/prev, case · whole-cell · regex · column scope
`Enter` / `Esc` in the filter box	apply / clear the filter
click a header	sort (ascending → descending → off)
right-click a header	statistics · sort · filter by · number format · color scale · text/number · hide
right-click a cell	copy (with header / row #) · filter by this value · statistics
glyph: hover / click / drag / double-click	read · filter · brush a range · statistics
drag a column border / double-click it	resize / autofit
kind badge (top right)	change how the file is read
row # box	jump to a row
`Ctrl+C`	copy the selection as TSV

The filter language

Comparisons `= != > >= < <=`, boolean `and or not` (C-style `== &&` also work), `between a and b`, `in (…)`, `contains` / `like` / `matches`, `is blank` / `is filled`, functions `round int abs sqrt log exp pow min max clamp bin if coalesce ifnum isnum isnan`. Bare words are columns, quoted words are text; bracket awkward names: `["Cu (ppm)"] > 30`. The same expressions define calculated columns and the grade–tonnage factors.

lamina — part of the Auditable project, Geoscientific Chaos Union. Engine: `@gcu/lamina`; grid: `@gcu/loom`; expressions: `@gcu/expr`; archives: `fflate · fzstd · seek-bzip` (MIT). MIT.
Screenshots are generated from the app by an automated harness, so they track the current build. See the in-app **Help** menu for the terse quick reference.